

INF. II – mod.2 // Comp Intell

E1: modeling and simulation

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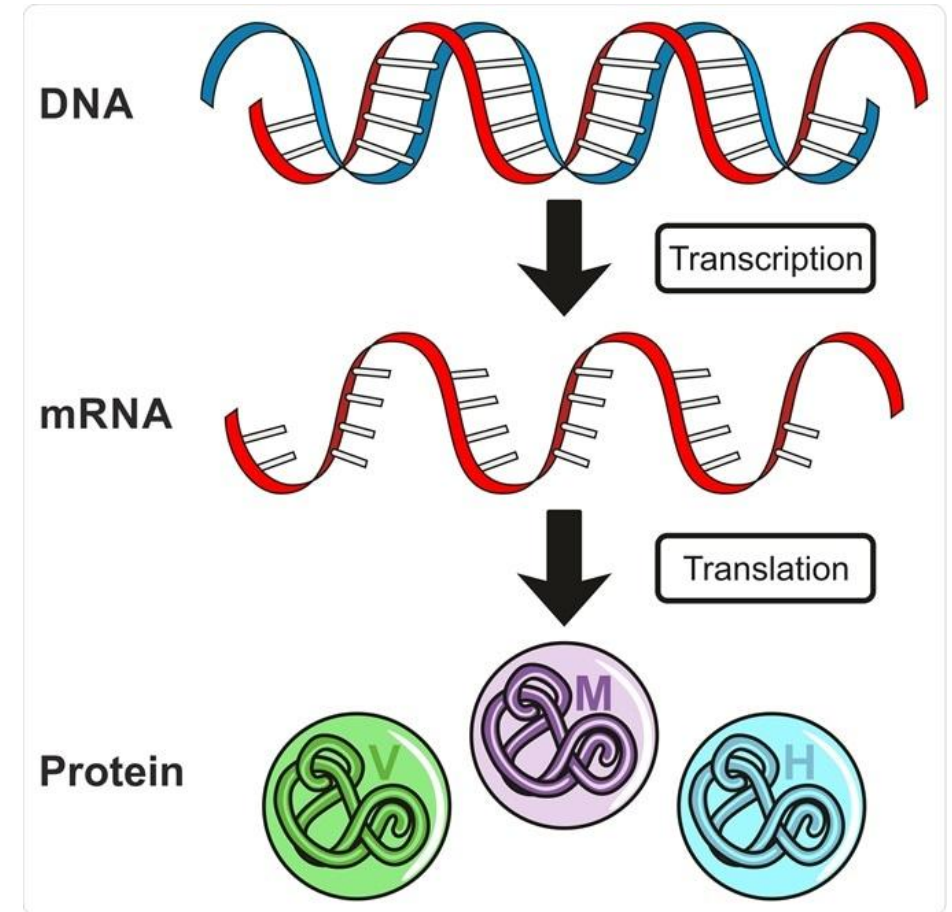
Outline



- The Central Dogma of molecular biology
- Modeling and simulation
- Discrete systems and the role of stochasticity
- Markov chains
- (Chemical) Master Equations
- A few words on random numbers
- Stochastic simulation

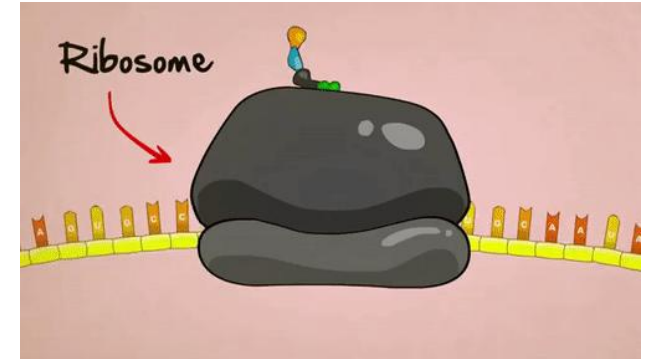
The Central Dogma

- The Central Dogma deals with the **flow of genetic information** in living beings
 - DNA is transcribed to mRNA
 - mRNA is translated to proteins
- There are several variations of the Dogma, but essentially all of them state that a **protein cannot be "translated back"** to DNA

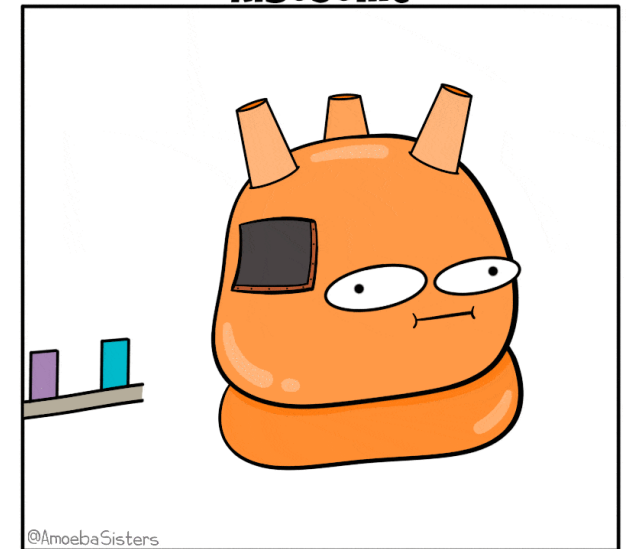


The role of the ribosome

- Ribosomes are **large macromolecular complexes**
- They are **responsible for translating an RNA sequence** (composed of a sequence of nucleotides – adenine, cytosine, guanine, uracil) **into a sequence of amino acids**
- The sequence of amino acids **folds to form a protein**
- **The shape of the protein determines its function**



RIBosome

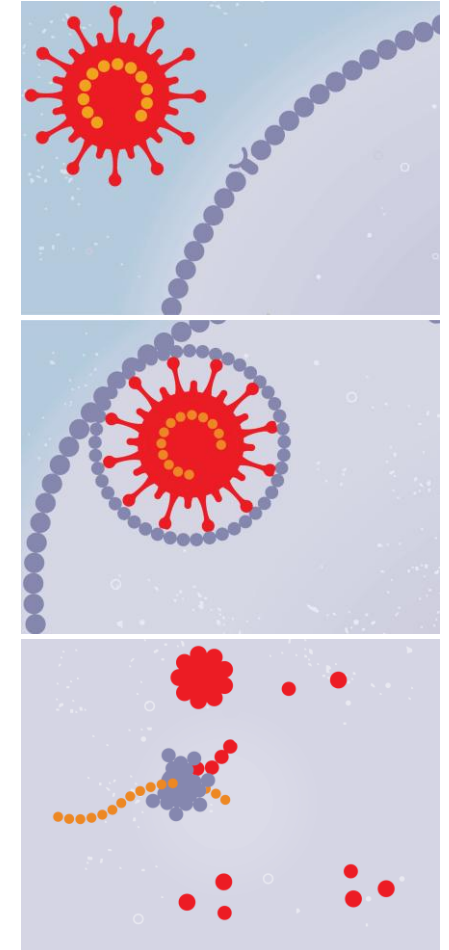


Protein synthesizers of the cell

Gene expression, proteins, and diseases



- The conversion of DNA into proteins is called **gene expression**
- Many **diseases** are caused by errors in the DNA (e.g., random mutations)
 - In turn, this leads to a **wrong sequence of amino acids**
 - A wrong sequence of amino acids implies a **different folding** of the protein
 - A different folding yields a **different function**, or even **no function at all**
 - E.g., cancer is often caused by mutations in DNA
- There can also be a **malicious interference** with the gene expression activity
 - Viruses generally introduce **genetic material** into the cell
 - Ribosomes **cannot know** that the incoming RNA is exogenous so they begin translation
 - It is hence important to **study and control these mechanisms**

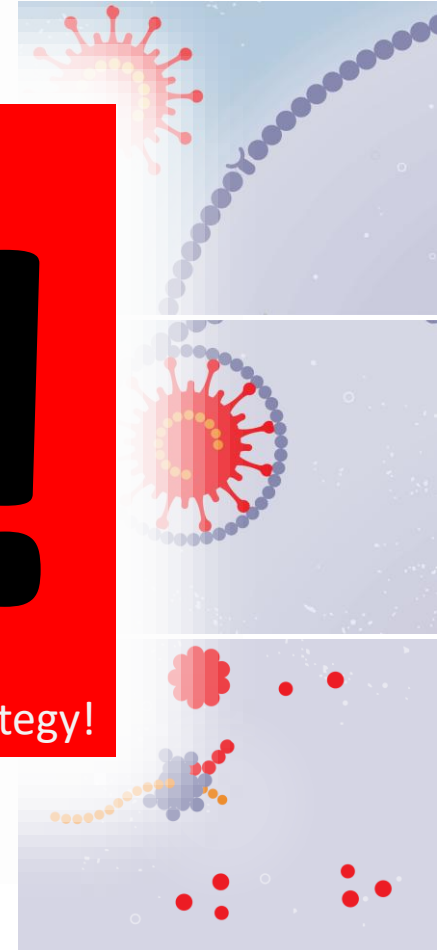


Gene expression, proteins, and diseases

- The conversion of DNA to RNA and then to proteins
- Many diseases are caused by errors in these processes
 - In turn, these errors can be caused by
 - A wrong sequence of the DNA
 - A different rate of transcription or translation
 - E.g., cancer
- There can be external factors that interfere with these processes
 - Viruses
 - Ribosomes **cannot know** that the incoming RNA is exogenous and begin translation
 - It is hence important to **study and control these mechanisms**

HOW?!

Feel free to suggest a strategy!



Modeling and simulation



- When we are facing a problem in a complex system, we can exploit **computers**
 - Cheaper and "faster" than conventional laboratory experiments
- In particular, **we can create a model** of the complex system, describing its **behavior**, **calibrate** its functioning against **experimental data**, and finally **simulate** any scenario we want to study
- We might be tempted to **simulate all physical processes** involving gene expression: it cannot get more accurate than this!

Some basic terminology



MODEL

A **rigorous** and **unambiguous description** of a system, precisely defined according to an adequate **mathematical formalism** and to a proper **level of abstraction**

SIMULATION

A computer-based way to **determine any emergent features** and/or to **predict** the temporal evolution of the system

EMERGENT PROPERTY / BEHAVIOUR

Any **structural** or **dynamical feature** of a system that cannot be understood/explained by the simple enumeration of its components, but can only be derived (i.e., **it emerges**) as a consequence of the interactions among the system components

COMPUTATIONAL METHOD

An **automatic solution** (based on a specific algorithm) used to **solve hard problems** and to **analyze** the system in different conditions

Hard truths about models



Essentially, all models are wrong, but some are useful.

(George E.P. Box)

In fact, all modelers should be prepared to answer the question:
“what do you know now that you did not know before?”.
If the answer is “that I was correct”, it is best to look elsewhere.

(J.M.Bower & H. Bolouri (eds.), The MIT Press, 2001)

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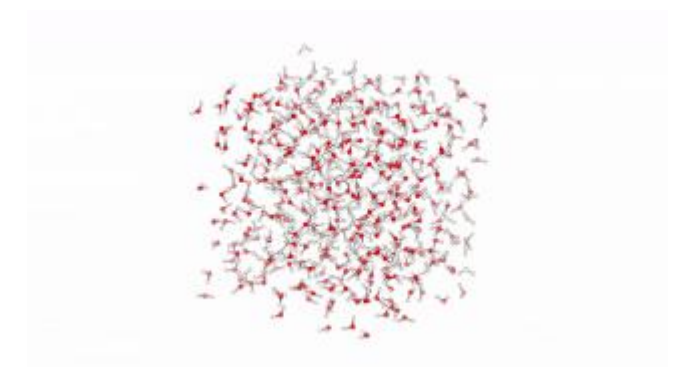
Models are always wrong, but sometimes they are also dangerous.

(Me, right now)

Molecular Dynamics (MD)

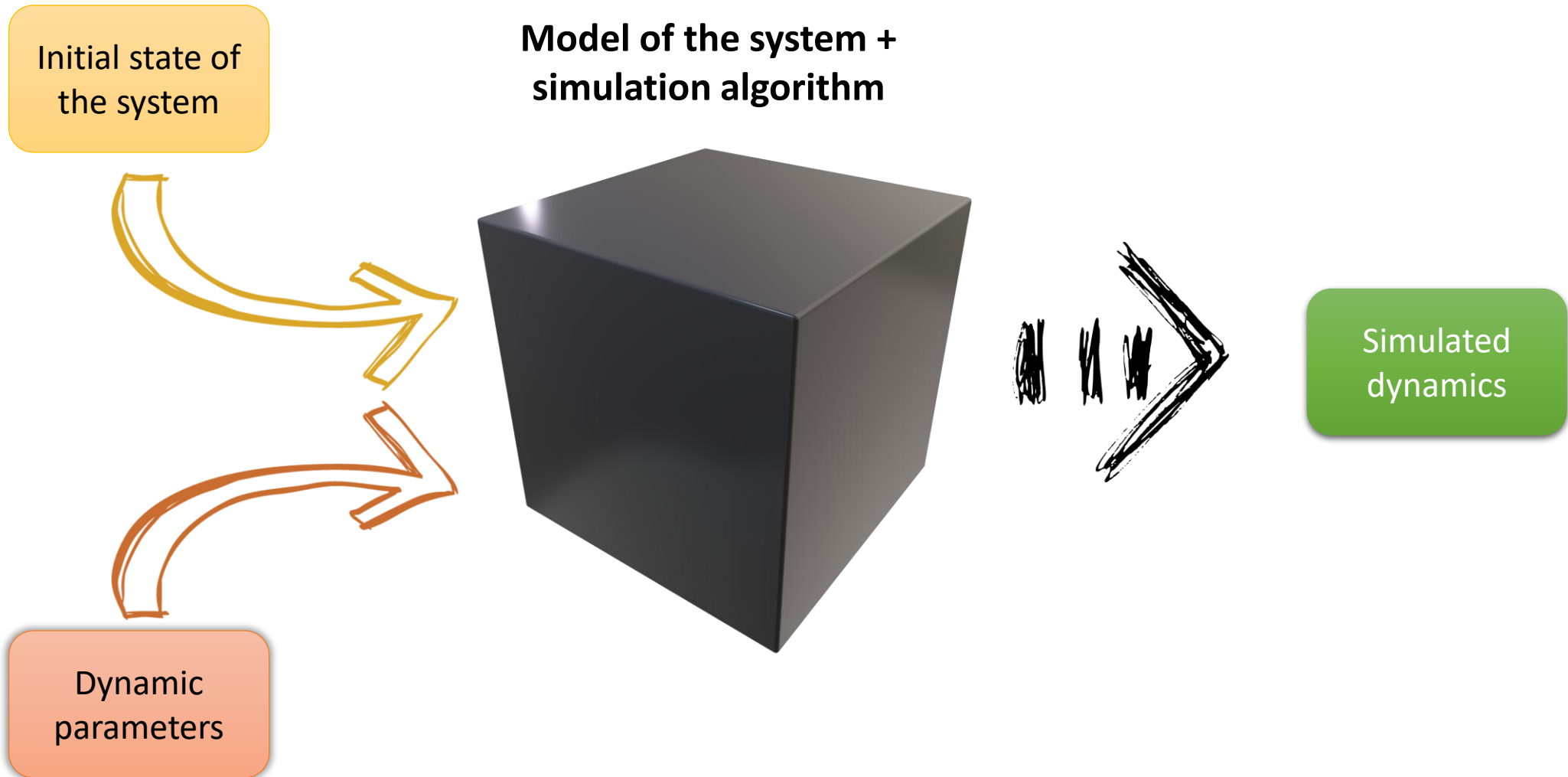


- MD is a computer simulation method based on calculating the trajectory of **of atoms of molecules** in a 3D space
 - That is, discretize time and calculate (step-by-step) the positions, velocities, and acceleration of each atom
- Atoms interact according to inter-atomic **force fields**
- Analytical solutions for the equations of motion are clearly **impossible**, so we need to use **numerical methods**

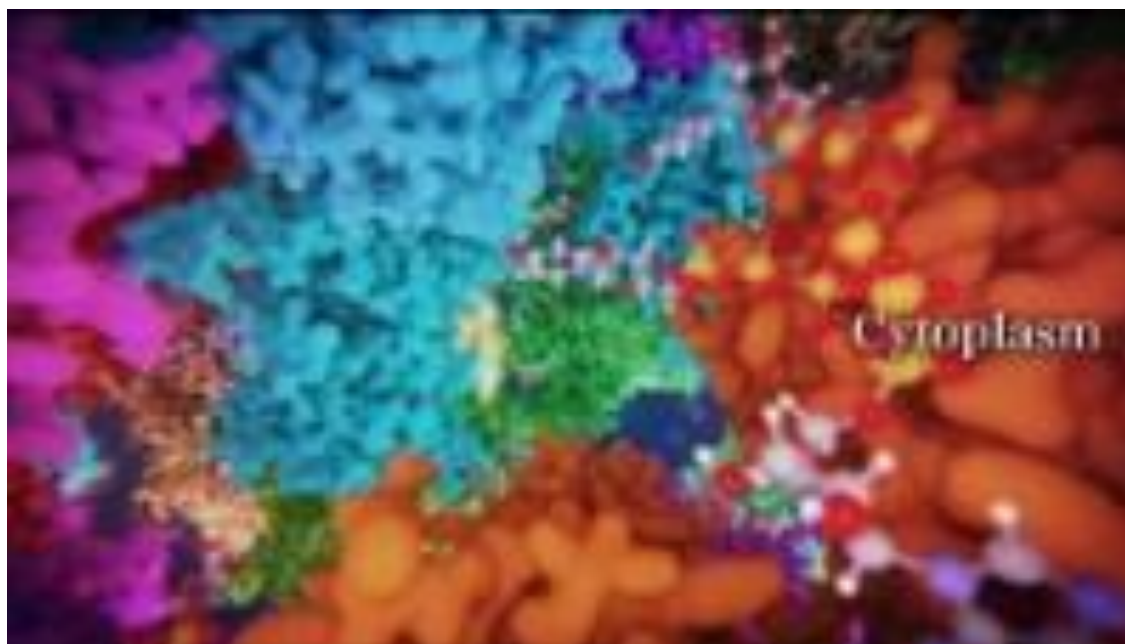


Water, 100 ps simulation

Abstract vision of modeling/simulation (reprise)



Molecular Dynamics (MD)



<https://www.youtube.com/watch?v=5JcFgj2gHx8>

3. Models of cellular environments

3.1. Fully atomistic models

The most detailed models of cellular environments consist of fully atomistic descriptions of biological macromolecules, metabolites, and the surrounding solvent and/or membrane environment (see Fig. 1). Such models have 0.1 nm spatial resolution and fs time resolution if they are investigated via molecular dynamics (MD) simulations. Because of prohibitive computational costs, fully atomistic MD simulations were applied up until recently only to the study of single macromolecules at a time and rarely exceed microsecond time scales. In

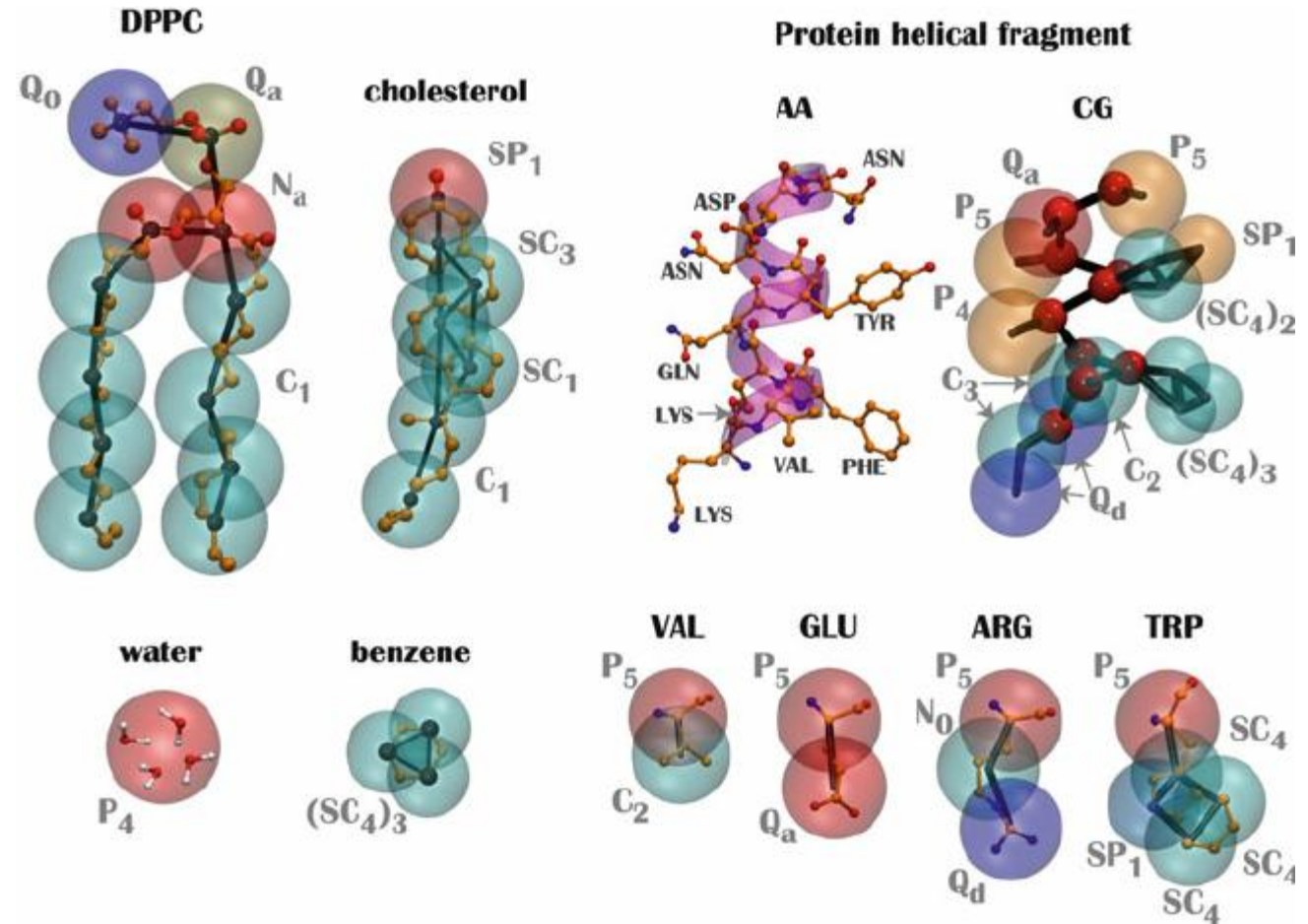
biologically accurate cellular models. A major challenge in developing such models is the need to cover a wide range of scales. Atomistic details of molecular processes occur on length scales of 0.1 nm while cellular dimensions are between 300 nm for the smallest bacterial cells and 100 μm for large eukaryotic cells. Biologically relevant time scales range from nanosecond to microsecond time scales for the internal dynamics of individual molecules to time scales of seconds to hours for entire biological processes. While a model that represents an entire cell in full atomistic detail is conceivable, it would be extremely costly to reach nanosecond time scales with today's most advanced computers and impossible to cover seconds for such a system in the foreseeable future. Instead, the development of reduced and multi-scale models is the only viable strategy today for developing physically

[Feig and Sugita, J Mol Graph Model, 2013]

Coarse-grained models



- **MARTINI** "coarse-grained" force fields
- Forces of multiple atoms modeled as **beads**
- On average, **four heavy atoms** and associated hydrogens are represented by a **single interaction center**

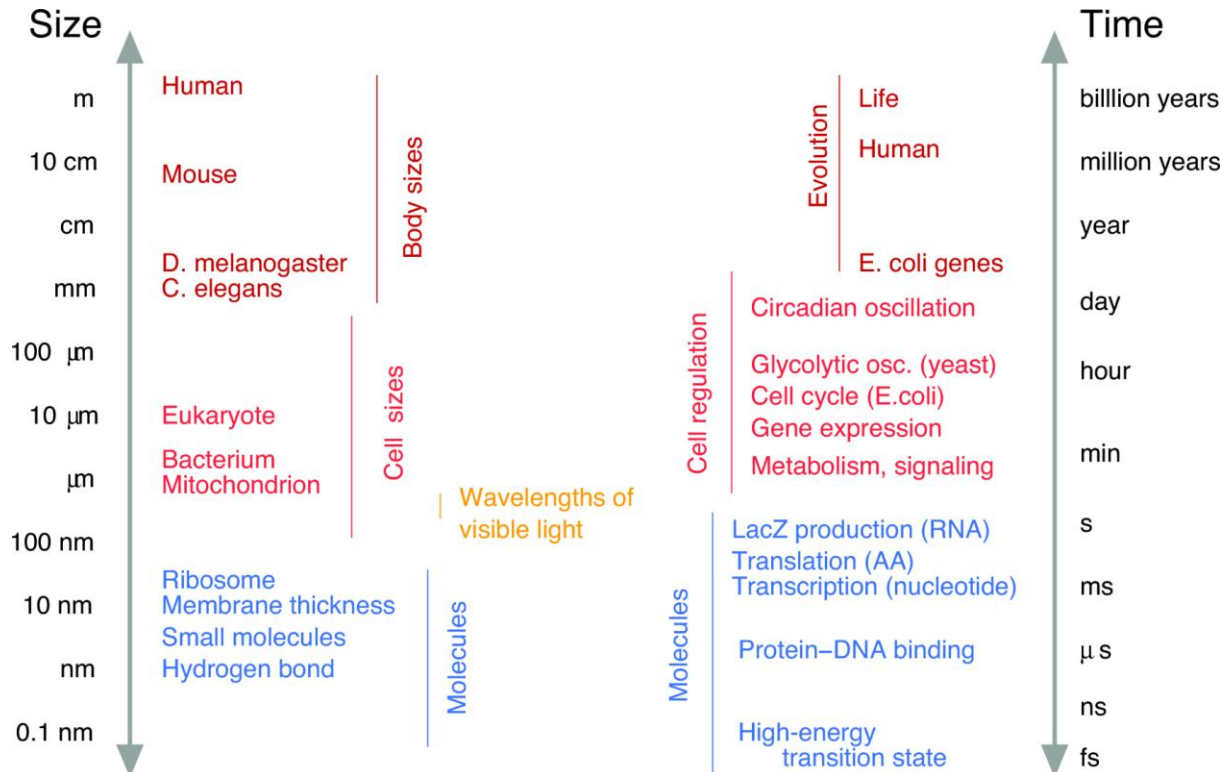




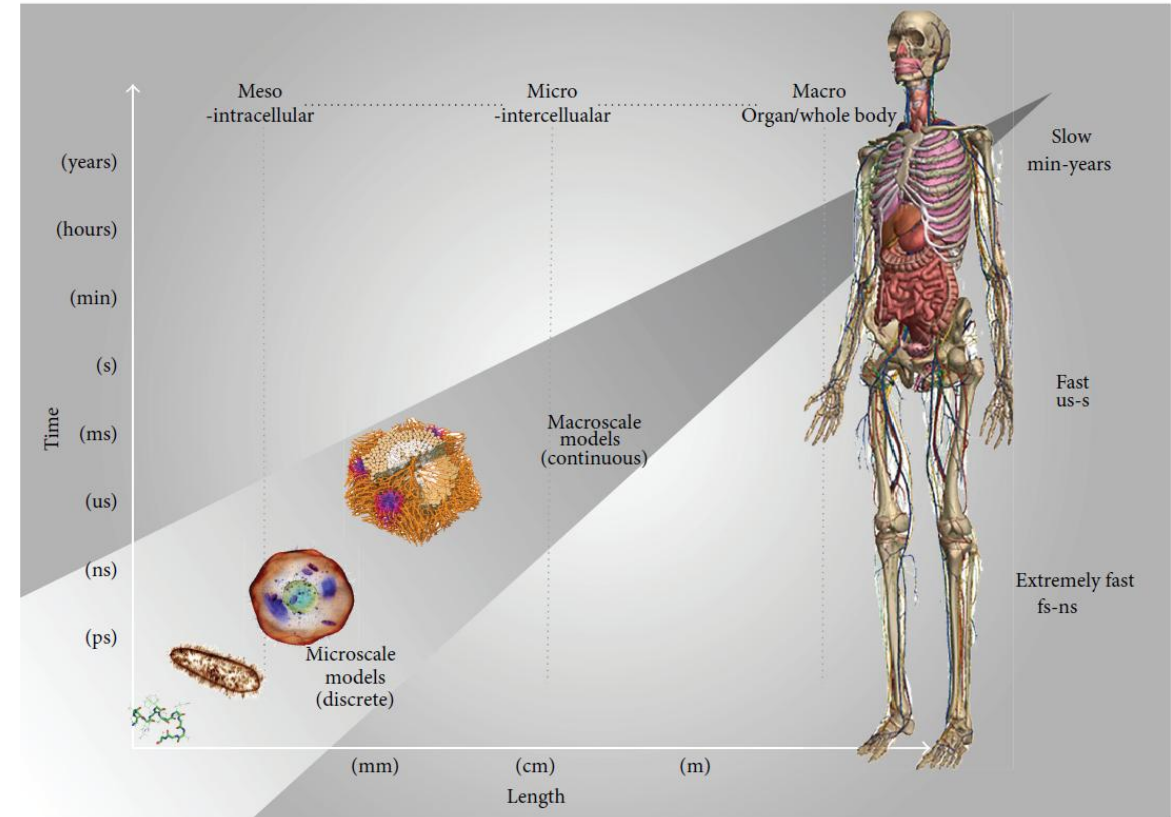
Scaling up: reaction-based modeling

- MD is powerful but **computationally demanding** (even coarse-grained)
 - Still largely used for **drug design** (e.g., docking) or **repurposing** (see, e.g., [Sacco *et al.*, *Front Mol Biosci* 2021])
 - **Coarse-grained** simulation can be used for **structural analyses** of polymers and polypeptides (see, e.g., [Nobile *et al.*, *Scientific Reports* 2021])
 - Still, the **time-scale of MD** is $\approx 10^{-12}$ s: too small to investigate many phenomena
 - **We do not necessarily need such level of detail**
 - Also, the exact position of atoms might be overkill: perhaps, knowing the **amounts of the items** in the system is **enough to draw conclusions** about its behavior
- **There is no single modeling approach that spans simultaneously all biological scales**
- However, we are often interested in investigating the scale of **(bio)chemical reactions**, that is, in the **order of seconds or minutes**
 - Simulating minutes of (bio)chemical reactions with MD – even using the most powerful machines in the planet – would require centuries!

Temporal and spatial scales



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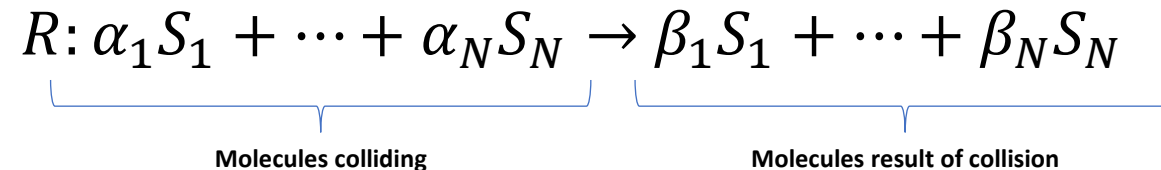


[Castiglione *et al.*, *BioMed Res Int*, 2014]



Reaction-based models

- In general, a chemical reaction can be seen as the **collision between two or more reactant molecules**
- Given a set $\mathcal{S}$ of N chemical species, we can define a **general reaction** R as the following **transformation**:



- The chemical species appearing before and after the " $\rightarrow$ " are named **reactants** and **products**, respectively
- α_i and β_i are the **stoichiometric** coefficients of species S_i (i.e., the number of molecules) involved in the reaction as reactant or product, respectively
 - They can be also 0 (=not involved)



Reaction-based models: basic definition

- Given a (biological) system with any kind of **interacting components**, a formalization of its components and their interactions can be done by specifying:
 - A set of (molecular) species $S_1, \dots, S_N$
 - A set of (biochemical) reactions $R_1, \dots, R_M$
 - The initial amounts $X_1, \dots, X_N$ of species $S_1, \dots, S_N$, respectively
 - The (kinetic) constants $k_1, \dots, k_M$ characterizing the chemical/physical properties of reactions $R_1, \dots, R_M$, respectively
- NB: A **volume** Ω (where reactions take place) is usually given, and assumed to be invariable and **well-stirred** (i.e., species are uniformly distributed inside Ω)



What do we mean for species amount?

- In reaction-based models, species **amounts** can be given either as **number of molecules** (integer numbers) or as **concentrations** (real numbers)
- The way species amounts are specified depends on how we want to «interpret» the model:
 - As a stochastic model → **number of molecules**
 - As a deterministic model → **concentrations**
- FYI, you can transform number of molecules into concentrations and viceversa:
 - (number of molecules of species S_i) = (concentration of S_i) $\cdot N_A \cdot V$
 - where N_A is **Avogadro's** number and V is the volume



Law of mass action

- According to the **law of mass action**, the **rate** of a chemical reaction is **proportional to the product of the number of molecules** of its reactants
 - In particular, the quantities are raised to a power equal to the coefficient that occurs in the chemical equation
- In the case of the elementary reaction $A+B \rightarrow *$ the mass-action kinetics leads to the following **rate equation**: $rate = k[A][B]$ where
 - $[A]$ and $[B]$ are the **concentrations** of the chemical species A and B, respectively
 - k is the **kinetic constant**
- In the case of the elementary reaction $A \rightarrow *$ the mass-action kinetics leads to the following **rate equation**: $rate = k[A]$

From reactions to differential equations



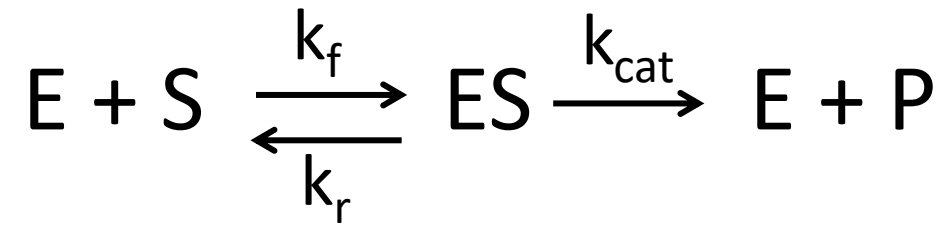
- Each **rate equation** is a **differential equation** that describes the **variation** over time of the **concentration** of any molecular species X , taking into account all its interactions with other species
 - Some terms will be **negative** as they represent processes that **reduce** the concentration (e.g., decomposition, phosphorylation, binding to other molecules)
 - Some terms will be **positive** as they represent processes that **increase** the concentration (e.g., synthesis, dephosphorylation, release, disassembly)

$$\frac{dX}{dt} = + \text{synthesis} - \text{decomposition} - \text{phosphorylation} + \text{dephosphorylation} - \text{binding} + \text{release} \dots$$

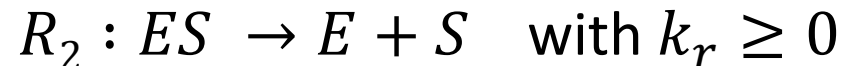
Example: enzyme kinetics



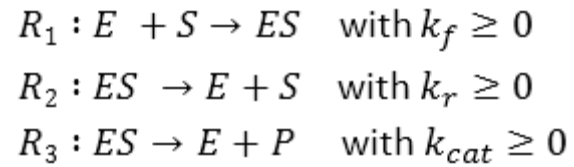
- Let's derive the system of ODEs corresponding to the following reactions:



- E (enzyme), S (substrate), ES (enzyme-substrate complex), P (product)
- Reactions:



Solution: enzyme kinetics



$$\frac{d[E]}{dt} = -k_f[E][S] + k_r[ES] + k_{cat}[ES]$$

$$\frac{d[S]}{dt} = -k_f[E][S] + k_r[ES]$$

$$\frac{d[ES]}{dt} = k_f[E][S] - k_r[ES] - k_{cat}[ES]$$

$$\frac{d[P]}{dt} = k_{cat}[ES]$$



From reactions to ordinary differential equations

Reaction Syntax	ODEs
$S_1 \xrightarrow{k} S_2$	$\begin{aligned} \dot{[S_1]} &= -k[S_1] \\ \dot{[S_2]} &= k[S_1] \end{aligned}$
$S_1 + S_2 \xrightarrow{k} S_3$	$\begin{aligned} \dot{[S_1]} &= -k[S_1][S_2] \\ \dot{[S_2]} &= -k[S_1][S_2] \\ \dot{[S_3]} &= k[S_1][S_2] \end{aligned}$
$S_1 + lS_2 \xrightarrow{k} S_3$	$\begin{aligned} \dot{[S_1]} &= -k[S_1][S_2]^l \\ \dot{[S_2]} &= -kl[S_1][S_2]^l \\ \dot{[S_3]} &= k[S_1][S_2]^l \end{aligned}$
$S + S \xrightarrow{k} \dots$	$\dot{[S]} = -2k[S]^2$
$S_1 + S_2 \xrightarrow{k} S_3 + S_3$	$\dot{[S_3]} = 2k[S_1][S_2]$
$S_1 \xrightleftharpoons[k_2]{k_1} S_2$	$\begin{aligned} \dot{[S_1]} &= -k_1[S_1] + k_2[S_2] \\ \dot{[S_2]} &= k_1[S_1] - k_2[S_2] \end{aligned}$
$S_1 + S_2 \xrightleftharpoons[k_2]{k_1} S_3$	$\begin{aligned} \dot{[S_1]} &= -k_1[S_1][S_2] + k_2[S_3] \\ \dot{[S_2]} &= -k_1[S_1][S_2] + k_2[S_3] \\ \dot{[S_3]} &= k_1[S_1][S_2] - k_2[S_3] \end{aligned}$
$\emptyset \xrightarrow{k} S$	$\dot{[S]} = k$
$S \xrightarrow{k} \emptyset$	$\dot{[S]} = -k[S]$

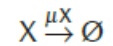
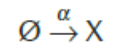
These equations directly derive from the mass-action law

We will get back to rate equations in future lectures about differential equations

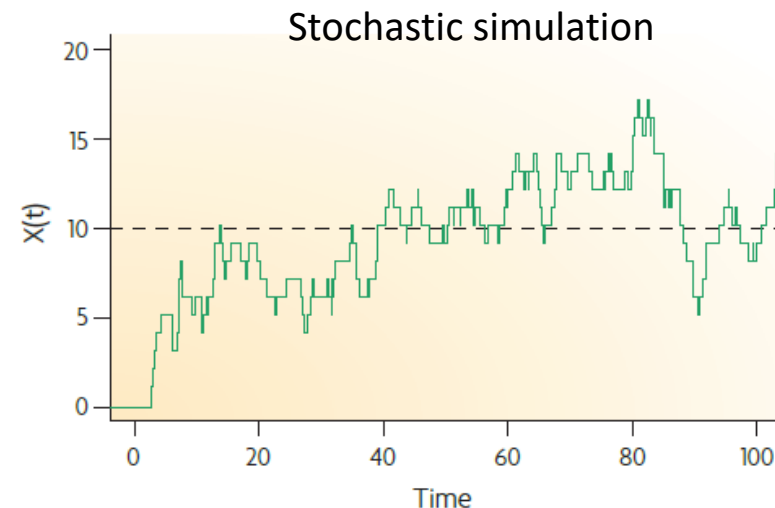
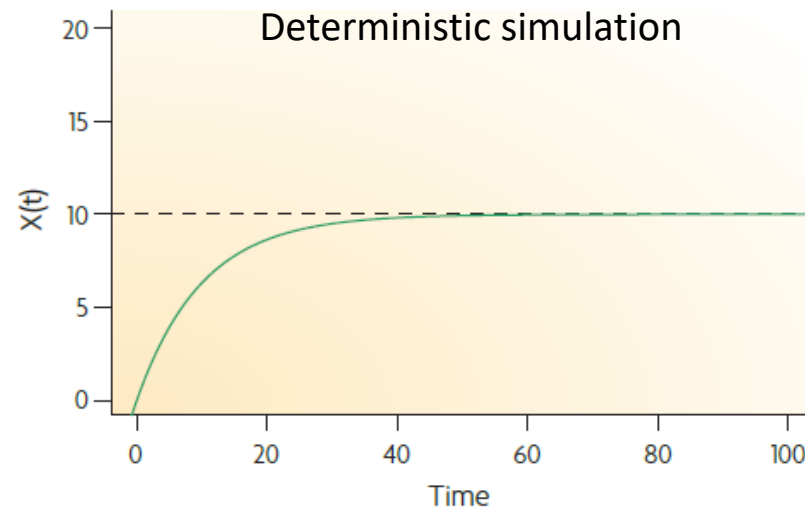
Stochastic vs deterministic approach

Box 1 | A simple model for protein production and degradation

Consider the following artificial model for production and degradation of a single protein, X : the protein is produced at a constant rate α , and each protein molecule is independently degraded at a constant rate μ . This can be written using chemical reaction notation as:



Let the number of molecules at time t be denoted X_t , and assume that there are initially no protein molecules, so that $X_0 = 0$. The plots show the case $\alpha = 1$, $\mu = 0.1$. The parameters are purely illustrative and not intended to model any real biological system.

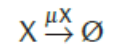
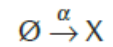


Wilkinson *Nature Reviews Genetics* 10, 2009

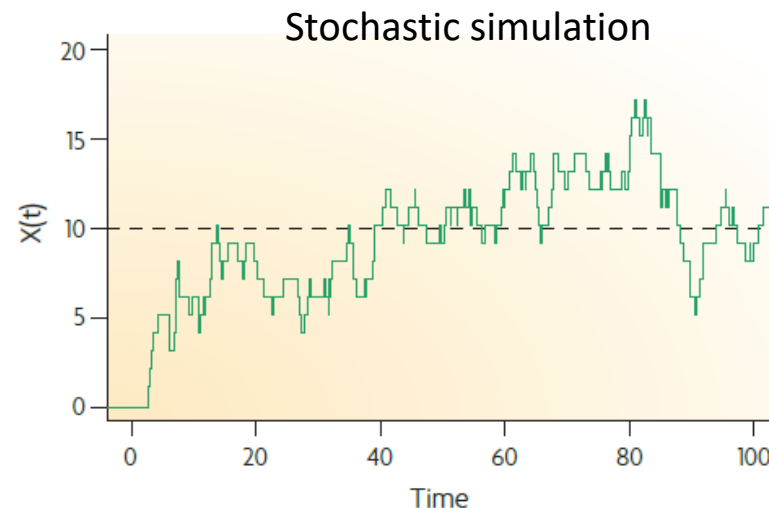
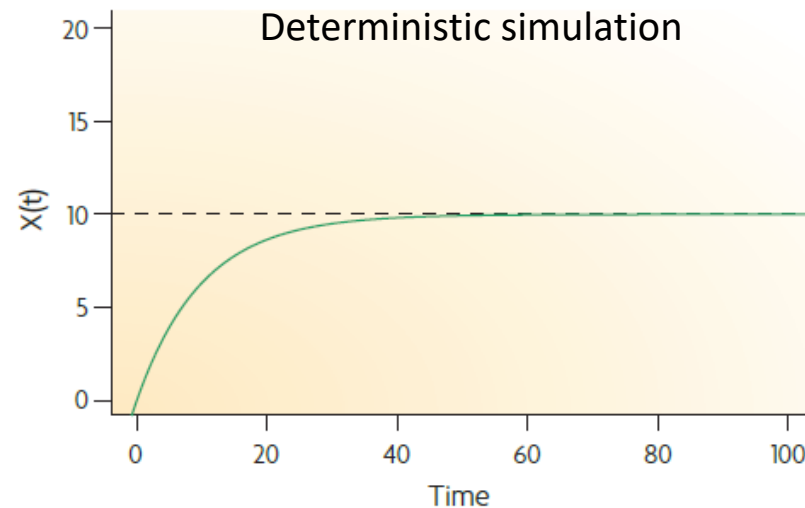
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Wilkinson *Nature Reviews Genetics* 10, 2009

We will now
spend some
time time
discussing
stochastic
simulation

What's biological noise?



- **Cell behavior** and the **extra-cellular environment** are **stochastic**
 - The genetic circuits that regulate cellular functions are subject to stochastic fluctuations, or **noise**, in the levels of their components
- **Cell phenotypes vary** across isogenic populations and in individual cells over time
- Gene expression measurements in single cells revealed stochastic bursts of both mRNA and protein synthesis in many organisms → **intracellular heterogeneity**
- Stochasticity represents a **beneficial mean** exploited by living organisms to respond to ever-changing environments by adapting their behaviors
- Cells that respond stochastically can out-compete cells that are overcome by environmental changes → at the longest timescales, noise may facilitate **evolutionary transitions**



Noise: bursts and propagation

- Many biochemical processes involve low molecule numbers or infrequent interactions, giving rise to stochastic fluctuations
- **Gene expression** is the best studied example of this phenomenon as it is central to almost all cellular functions, and depends on the low copy number of genes (1-2 molecules per cell)

Concept of burst

- Gene promoters can stochastically switch between 'off' and 'on' states, resulting in bursts of mRNA production
- Each individual messenger RNA is typically translated many times to produce many proteins, generating corresponding protein bursts

Concept of propagation

- The rates of gene expression are influenced directly by the levels of transcription factors and other upstream components, which are themselves subject to bursting and time averaging
- As a result, fluctuations in the expression of one gene propagate to generate fluctuations in downstream genes

Effects of biological noise

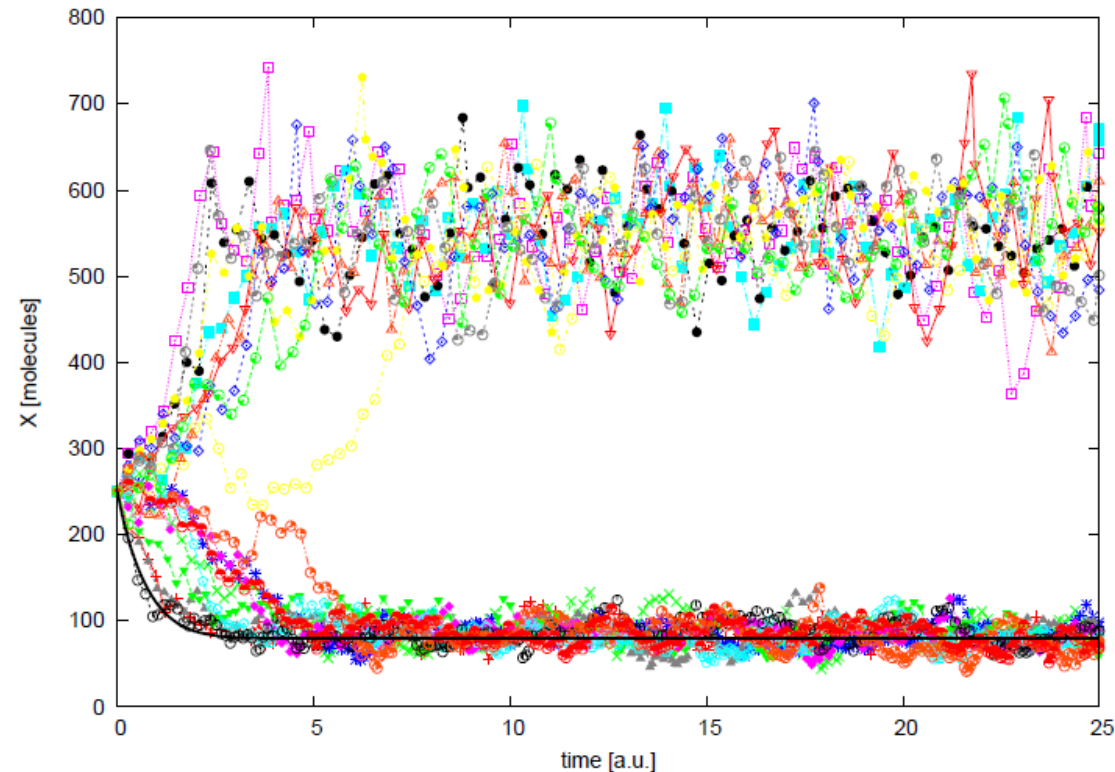


- Stochastic fluctuations at the molecular level can induce different *macroscopic behaviors* at the cellular level:
 - **Bistability**
 - **Switching** phenomena

Example: the Schlögl model



- Deterministic vs stochastic simulations ($X(0)$ is always 250 molecules)

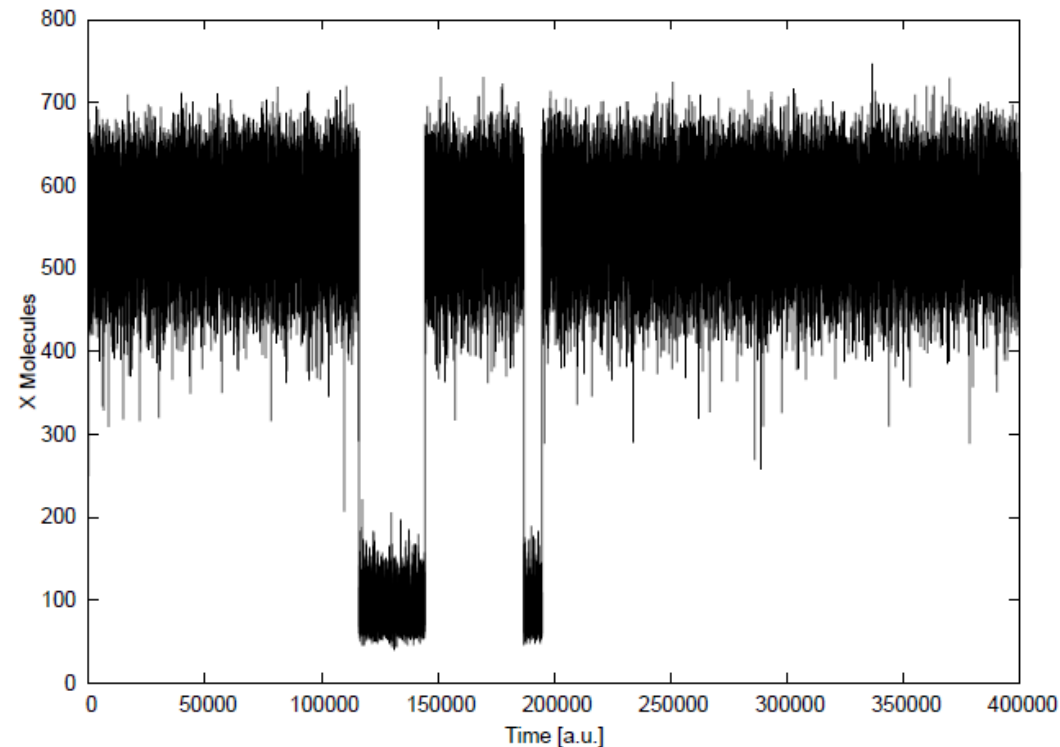


Black line =
ODE!

State switching in the Schlögl model



- Stochastic transitions between steady states



Stochastic > deterministic



- Stochastic modeling and simulation gives a **better characterization** of the system under investigation, reproducing the emergent effects
- **Professor Besozzi** (from University of Milano-Bicocca) will give a seminar about the study of **emergent phenomena in complex systems** at the end of the course (chaos and multistability in particular)





Markov chains

- A **Markov chain** (or process) is a **stochastic model describing a sequence of possible events**, in which the **probability of the events only** depend on the **last state of the system**
 - This property (of being completely "memoryless") is called **first order Markovian process**
 - A Markov chain can be either **discrete** or **continuous-time**
- A Markov chain is characterized by a (countable) **set of possible states**
 - **Countable**: each state can be biunivocally associated to one **natural number**
 - There can be one **initial state**
- The state of the system changes according to **transitions**
 - Transition leads from one state to another state according to a given **probability**

Stochastic simulations



- **Stochastic** simulations are simulations of systems whose **variables change randomly** according to some **probabilities**
 - Such probabilities are **calculated** according to the **current state of the system**
 - The probabilities are then used to **update the state** of the system
 - The simulation can either stop or **start again from the new state**
 - We will **define a "trajectory"** as the **sequence of states** generated by the simulation
- Given the fact that the change is stochastic (i.e., random), **two independent simulations of the same system can yield two completely different trajectories**
- In order to perform stochastic simulations, we must be able to **generate random numbers** in a computer. **How would you do that?**

"Random" sequence generators



- The only way to **generate *true* random numbers** is by leveraging some ***actually* random event**
 - That is, by sampling some **physical random event**, e.g., **radioactive decay** or measuring **qubits in superposition**
 - This might be **overkill** for our purposes
- **We might prefer to generate random numbers** on a computer
 - However, conventional computers are **deterministic** in nature
 - If we repeat the **same algorithm on the same input we get the same output** (hopefully...)
 - How can we **generate random numbers** using a deterministic algorithm? **Any ideas?**



Randomness



Anyone who attempts to generate random numbers by deterministic means is, of course, living in a state of sin.

— *John von Neumann* —

AZ QUOTES



Pseudo-random numbers

- **Pseudo-Random Number Generators (PRNGs)** are not random at all, but they use a deterministic process to **generate a sequence** of numbers which **appears to be statistically random**
- PRNGs generally start with an **initial value** (named "the **seed**") and apply some **complex formula** to generate the **next value** and iterate the process
- The formula must be clever and **generate all possible values that we might need**, and **possibly never enter a short loop** ("period") in which the PRNG generates the same numbers over and over
 - Let's discuss this topic with a simple example

A simple example: Linear Congruential Generators



- A **Linear Congruential Generator** (LCG) is an algorithm which yields a sequence of pseudo-random numbers calculated with a discontinuous piecewise linear equation. One possible LCG is:
 - Assume an initial value (i.e., seed) $S(0)$
 - The next value is calculated as $S(t + 1) = [a \cdot S(t) + b] \bmod m$
 - This equation generates a sequence of pseudo-random integer numbers in $[0, m)$
- Let's try with some values: $S(0)=15$, $a=2$, $b=3$, $m=16$
 - $S(1)=(2 \cdot 15 + 3) \% 16 = 1$
 - $S(2)=(2 \cdot 1 + 3) \% 16 = 5$
 - $S(3)=(2 \cdot 5 + 3) \% 16 = 13$
 - $S(4)=(2 \cdot 13 + 3) \% 16 = 13$ **OUCH!**

Not so simple linear congruential generators



- By carefully choosing the values for a , b and c we can **prevent getting stuck in fixed points** and also generate sequences with a **long period**
 - No matter what we do, when we generate a sequence of numbers choosing from a finite set, the sequence will always be periodic
- There is a theorem, named Hull–Dobell, that gives the requirements to generate a sequence with period m (maximal)
 - Example: values used by the linux C compiler (GCC): $m = 2^{31}$, $a = 1103515245$, $b = 12345$



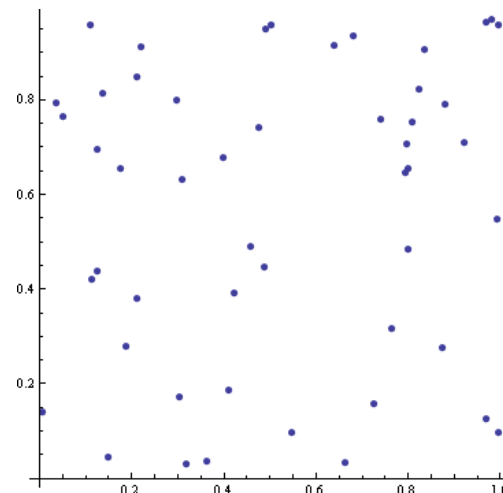
What about Python?

- **Python** (random module) uses a PRNG named **Mersenne Twister**
 - See [[Matsumoto and Nishimura, ACM Tran Model Comp Simul 1998](#)]
 - One of the most tested, respected and reliable generators
 - It produces 53-bit precision floats
 - Has an impressive period of $2^{19937} - 1$
- **NumPy** (random module) uses a more recent PRNG named **PCG-64**
 - **Permutation Congruential Generators**
 - **Extend and improve LCGs**
 - 128-bit Permutation Congruential Generator, period of 2^{128}
 - **Faster** and **beats** more statistical tests than MT
 - See [[O'Neill and Melissa, ACM Tran Math Soft 2014](#)]

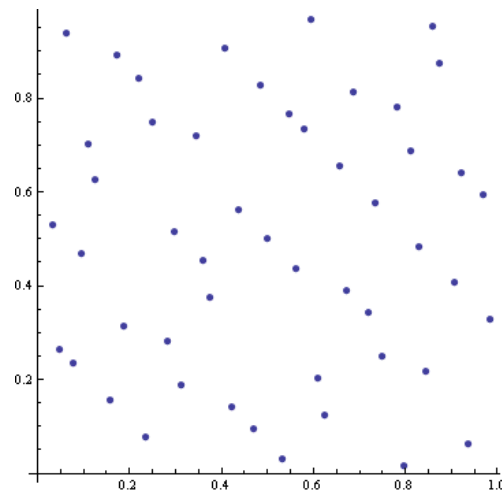
Quasi-random sequences

- Pseudo-random sequences try to generate random numbers that seem to **statistically random** and uncorrelated
 - However, true randomness is **not evenly distributed**
 - **Quasi-random sequences** (also called low-discrepancy) try to **cover the space more uniformly**
 - This is often a **desirable property for analyses** (more about this at the end of the course)

50 random points sampled
from uniform distribution

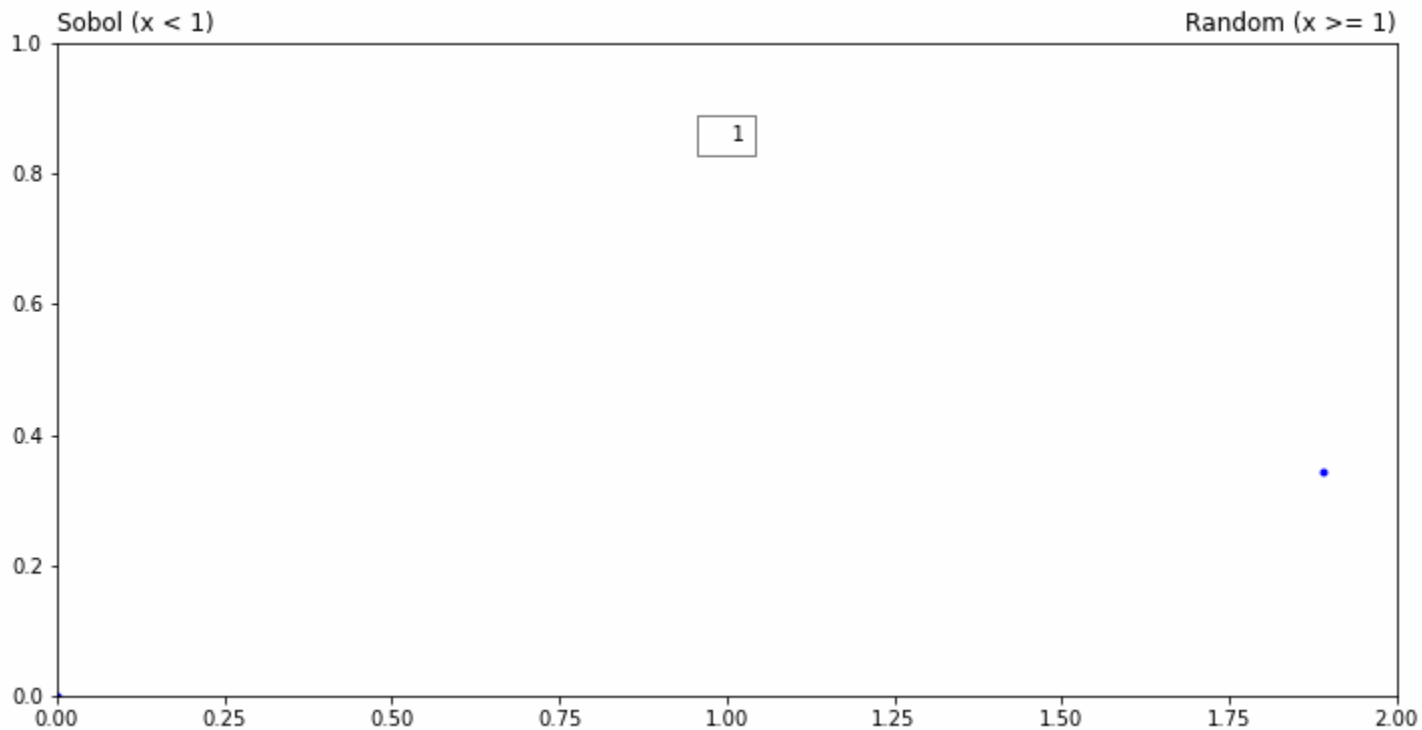


50 random points from a
quasi-random sequence



Images taken from John D. Cook's blog
<https://www.johndcook.com/blog/2009/03/16/quasi-random-sequences-in-art-and-integration/>

Comparison between pseudo- and quasi-random



How to simulate the effects of biological noise?



- Given a reaction R_μ and its (stochastic) kinetic constant c_μ , the **fundamental hypothesis of the stochastic formulation of chemical kinetics** states that:

$c_\mu dt = \textit{average probability}$ that any particular combination of molecules appearing as reactants in reaction R_μ will *react* accordingly, in the next infinitesimal time interval dt

The key concepts of stochastic simulation



- Given a set of reactions and the current system state, we want to answer **two questions**:
 - **When will the next reaction take place?**
 - That is, how long do we have to wait before the next reaction will occur somewhere inside the volume?
 - **Which reaction will actually take place?**
 - Important: the reaction having the highest probability to occur is *not necessarily* the one that will take place!

Basic notions of stochastic algorithms



- Given a reaction-based model, each reaction R_μ ($1 \leq \mu \leq M$) is characterized by:
 - The **state-change vector** $\rightarrow$ it denotes the variation of the system state due to (one) occurrence of the reaction
 - The **propensity function** $\rightarrow$ it is needed to calculate the probability that the reaction may occur, in the current system state



State-change vector

1. State-change vector:

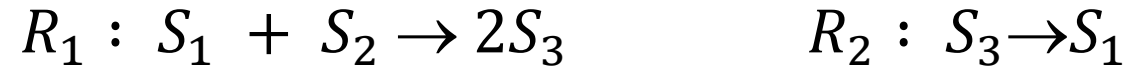
$$\mathbf{v}_\mu = (v_{1\mu}, \dots, v_{N\mu})$$

where $v_{i\mu}$ denotes the variation of the amount of species S_i , $1 \leq i \leq N$, due to the occurrence of reaction R_μ



State-change vector: examples

- Assume a reaction-based model consisting in three species S_1, S_2, S_3 and two reactions:



- The state-change vectors are $v_1 = (-1, -1, +2)$ and $v_2 = (+1, 0, -1)$
- Assume that the state at some time t is:

$$\mathbf{X}(t) = (X_1(t), X_2(t), X_3(t)) = (15, 7, 23)$$

- If we «simulate» the occurrence of reaction R_1 at time t we obtain:

$$\mathbf{X}(t+1) = \mathbf{X}(t) + v_1 = (15 - 1, 7 - 1, 23 + 2) = (14, 6, 25)$$

- If we «simulate» the occurrence of reaction R_2 at time $t+1$ we obtain:

$$\mathbf{X}(t+2) = \mathbf{X}(t+1) + v_2 = (14 + 1, 6 + 0, 25 - 1) = (15, 6, 24)$$



Propensity functions

2. The **propensity function** a_μ is defined as:

$$a_\mu(\mathbf{X}(t)) = c_\mu h_\mu(t)$$

where c_μ is the stochastic constant of reaction R_μ and $h_\mu(t)$ is a **combinatorial function** that characterizes the reaction (according both to its order and to the type of reactants)

- $a_\mu(\mathbf{X}(t))dt$ (i.e., $c_\mu dt$ multiplied by $h_\mu(t)$) represents the **total number of the possible distinct combinations of the reactants** of R_μ occurring in volume V at time t , that is, it is equal to the probability that, given the system state $\mathbf{X}(t)$, *one* reaction of type R_μ will take place somewhere in V in the time interval $[t, t+dt)$

Combinatorial functions



* $\rightarrow$ reaction products, $\longrightarrow h_\mu = 1,$

$S_j \rightarrow$ reaction products, $\longrightarrow h_\mu = X_j,$

$S_j + S_k \rightarrow$ reaction products $(j \neq k), \longrightarrow h_\mu = X_j X_k,$

$2S_j \rightarrow$ reaction products, $\longrightarrow h_\mu = X_j(X_j - 1)/2,$

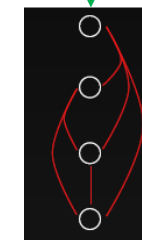
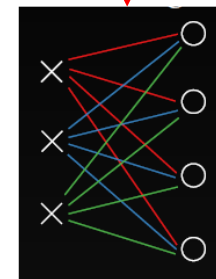
unlikely $S_i + S_j + S_k \rightarrow$ reaction products $(i \neq j \neq k \neq i) \longrightarrow h_\mu = X_i X_j X_k,$

unlikely $S_j + 2S_k \rightarrow$ reaction products $(j \neq k), \longrightarrow h_\mu = X_j X_k (X_k - 1)/2,$

unlikely $3S_j \rightarrow$ reaction products. $\longrightarrow h_\mu = X_j(X_j - 1)(X_j - 2)/6$

Example:

$R_1: S_1 + S_2 \rightarrow 2S_3$ and $X(t) = (3, 7, 5)$, then:
 $h_1 = 3 \cdot 7 = 21$ and $a_1 = 21 \cdot c_1$





The Chemical Master Equation (CME)

- Since we are describing the system in probabilistic terms, we are interested in determining $P(\mathbf{X}, t | \mathbf{X}_0, t_0)$, that is, the probability to be in state $\mathbf{X}$ at time t , given the initial condition $\mathbf{X}_0$ at time t_0
- The **(Chemical) Master Equation (CME)** is the equation that determines the probability that each species will have a specified molecular amount at any given future time:

$$\frac{\partial P(\mathbf{X}, t | \mathbf{X}_0, t_0)}{\partial t} = \sum_{\mu=1}^M [a_{\mu}(\mathbf{X} - \mathbf{v}_{\mu})P(\mathbf{X} - \mathbf{v}_{\mu}, t | \mathbf{X}_0, t_0) - a_{\mu}(\mathbf{X})P(\mathbf{X}, t | \mathbf{X}_0, t_0)]$$

- a_{μ} is the propensity function of reaction R_{μ}
- $\mathbf{v}_{\mu}$ is the state-change vector of reaction R_{μ}

CME: an «unfeasible» approach

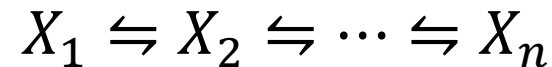


- The CME is actually a system of coupled ODEs, with **one equation for every possible state** in terms of reactant molecules
 - How many different states can a system have?



CME: an «unfeasible» approach

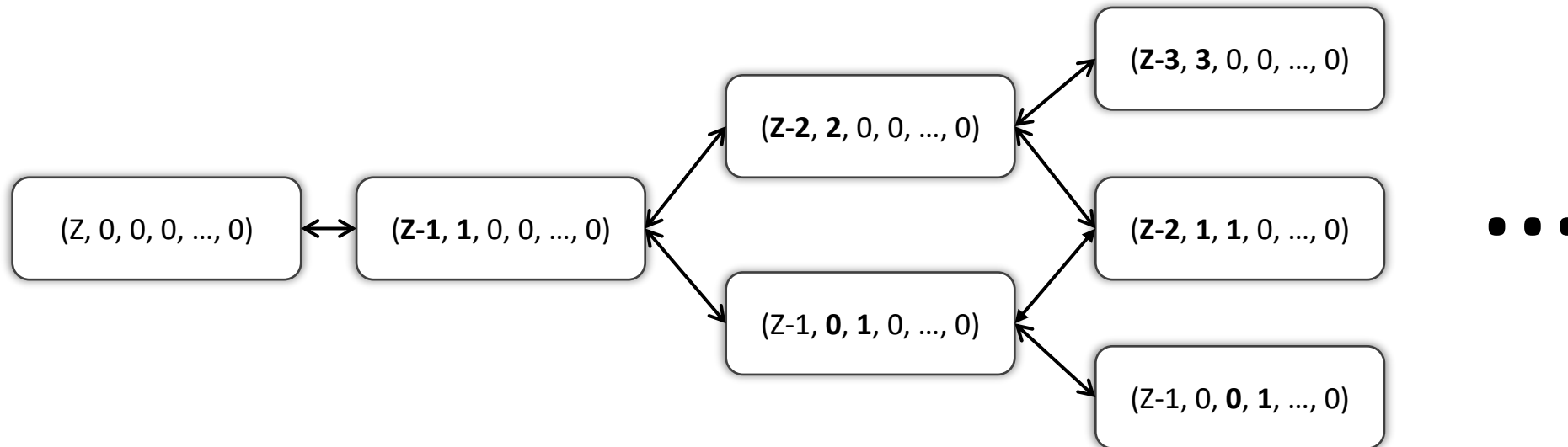
- The CME is actually a system of coupled ODEs, with **one equation for every possible state** in terms of reactant molecules
 - How many different states can a system have?
- Example: let's consider a chain of simple isomerization reactions



and a given initial state $\mathbf{X} = (Z, 0, \dots, 0)$

Possible states?

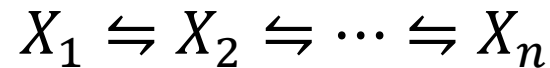
We must calculate the probability *for all possible states*





CME: an «unfeasible» approach

- The CME is actually a system of coupled ODEs, with **one equation for every possible state** in terms of reactant molecules
 - How many different states can a system have?
- Example: let's consider a chain of simple isomerization reactions



and a given initial state $\mathbf{X} = (Z, 0, \dots, 0)$. Then, the number of possible states is:

$$\frac{(Z + n - 1)!}{Z! (n - 1)!} \approx \left(\frac{eZ}{n - 1} \right)^{n-1}$$

The number of possible states is exponential!
 $O(Z^{n-1})$

- The CME **can be solved analytically for only a few simple cases**, and even numerical solutions are **prohibitively difficult** in real world scenarios



Another example: gene transcription

- The simplest Markov chain is a birth/death process, which well describes the dynamics of some genes
 - Only one species: mRNA
 - Two reactions: mRNA is **transcribed** ($\emptyset \xrightarrow{k} \text{mRNA}$), mRNA is **degraded** ($\text{mRNA} \xrightarrow{\gamma} \emptyset$)



Another example: gene transcription

- The simplest Markov chain is a birth/death process, which well describes the dynamics of some genes
 - Only one species: mRNA
 - Two reactions: mRNA is **transcribed** ($\emptyset \xrightarrow{k} \text{mRNA}$), mRNA is **degraded** ($\text{mRNA} \xrightarrow{\gamma} \emptyset$)
- By denoting with i the amount of mRNA molecules, the CME for this process is:
$$\frac{d}{dt}p_i(t) = -p_i(t)(k + \gamma i) + p_{i-1}(t)(k) + p_{i+1}(t)\gamma(i + 1)$$
- You can find the details of this CME in [\[Munsky et al., "Quantitative biology" 2018\]](#)
- Please focus your attention to the fact that **this CME has infinite possible states!**
Even it is a very small model, it is still untreatable

An alternative to CME: Gillespie's algorithm



- Instead of calculating the probability of future states using the CME, we can get *numerical realizations* of $\mathbf{X}(t)$
 - That is, *simulated trajectories* of the state of the system $\mathbf{X}(t)$ versus time t
- The key to generating simulated trajectories of $\mathbf{X}(t)$ is not the function $P(\mathbf{X}, t | \mathbf{X}_0, t_0)$, but rather a new **joint probability** function $P(\tau, \mu | \mathbf{x}, t)$, which is called the «reaction probability density function»
 - τ is the waiting time before the next reaction occurs
 - μ is the index of the next reaction R_μ
- **How can we calculate $P(\tau, \mu | \mathbf{x}, t)$ to perform a simulation?**

Gillespie's Stochastic Simulation Algorithm (SSA)



- Assuming a **well-stirred** system, **constant temperature and volume**, and strictly **mass-action kinetics** we can model and simulate a biochemical system as a **continuous-time, discrete-state Markov process**
- **Molecular counts** are positive integers that change by **discrete jumps** every time a reaction occurs
- This is, in a nutshell, **Gillespie's Stochastic Simulation algorithm (SSA)**
 - See [\[Gillespie, *J Phys Chem* 1977\]](#)

SSA

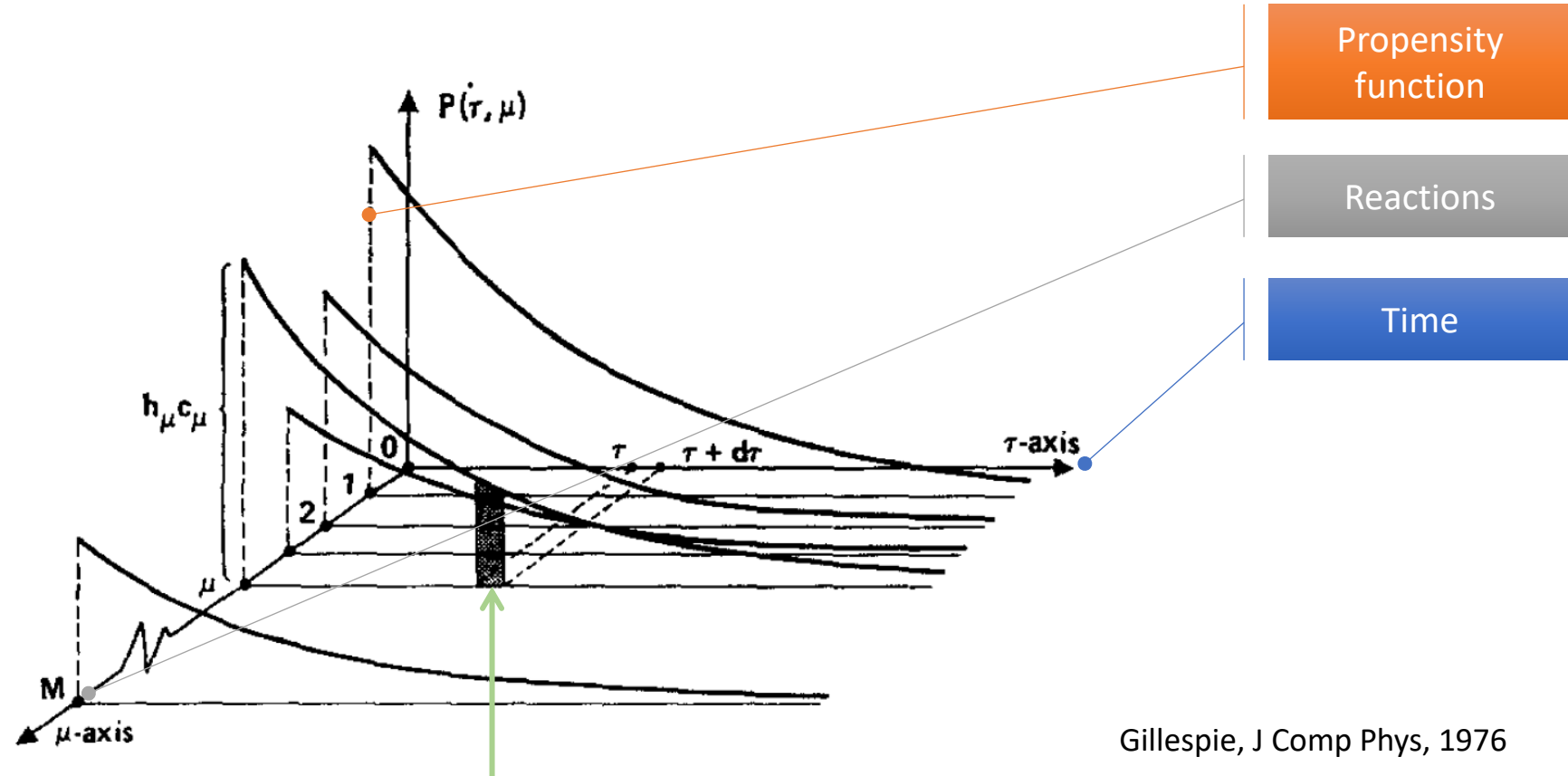


- The exact formula for $P(\tau, \mu | \mathbf{x}, t)$ is:

$$P(\tau, \mu | \mathbf{x}, t) = a_{\mu}(\mathbf{x}) \cdot \exp(-a_0(\mathbf{x})\tau)$$

where $a_0 = \sum_{\mu=1}^M a_{\mu}$

The joint probability of τ and μ



Gillespie, J Comp Phys, 1976

The shaded area (equal to $p(\tau, \mu)d\tau$) is the probability that, at time t , the next reaction in V will occur in the infinitesimal time $d\tau$ and will be a reaction of type R_μ

How to determine τ

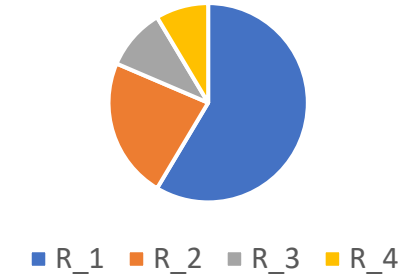
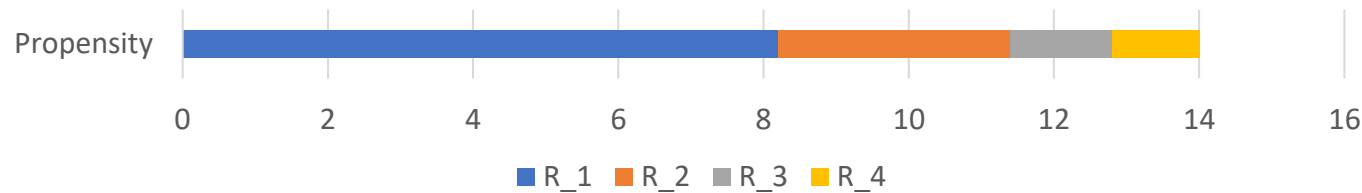
- We draw a random number r_1 from the uniform distribution in $[0,1)$ and calculate τ as:

$$\tau = \frac{1}{a_0(\mathbf{X})} \ln \frac{1}{r_1}$$

- Note that τ is **inversely proportional** to a_0
- Hence, a larger value of a_0 corresponds to shorter time steps $\tau \rightarrow$ the running time of the simulation will be higher

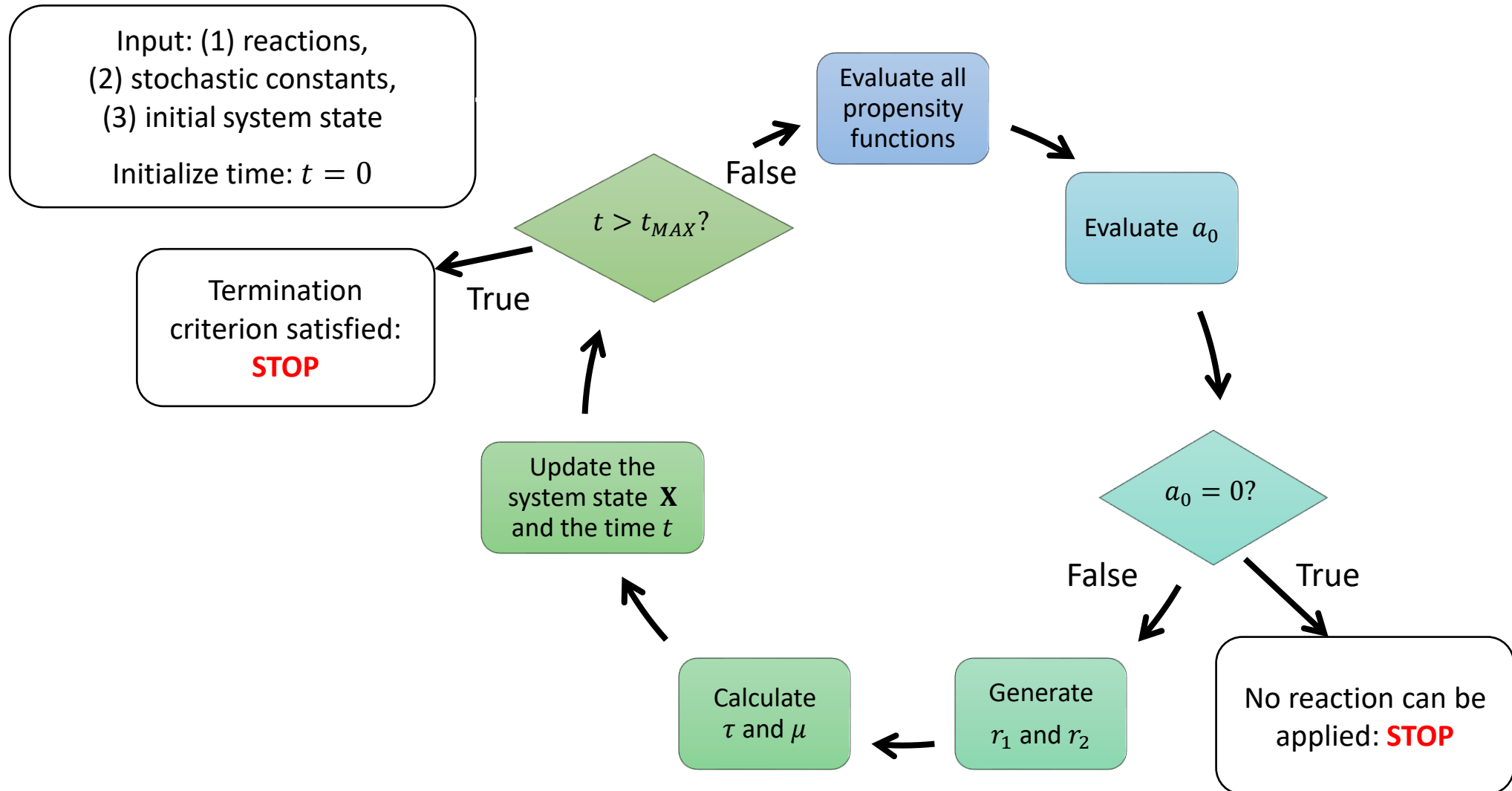
How to determine μ

- We draw another random number r_2 from the uniform distribution in $[0, a_0)$ and choose μ according to the following idea:



- Notice that the probability to apply reaction R_μ is **proportional** to a_μ
- Since $a_\mu = c_\mu \cdot h_\mu(t)$, it means that this probability is proportional to the stochastic constant multiplied by the number of reactant molecules currently present in the system
- Hence, if the system contains a large number of molecules $\rightarrow$ the running time of the simulation will be higher

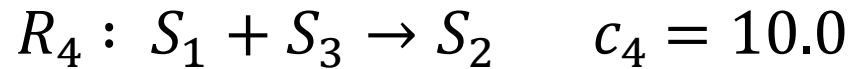
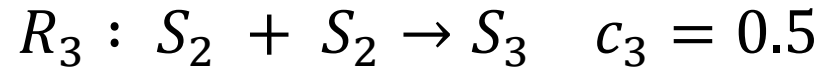
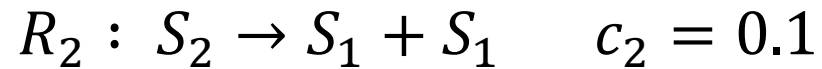
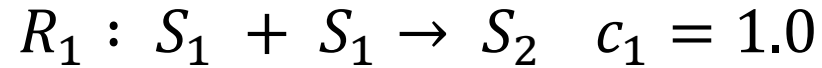
SSA - Flowchart



Let's try!



- Exercise! Consider the following set of reactions:



- The goal of this exercise is to simulate «by hand» the first 5 steps of the SSA, and see that different simulations yield different trajectories starting from the same initial state.
- In this exercise, let's assume that $\mathbf{X}(0) = (10, 0, 0)$

Hands-on

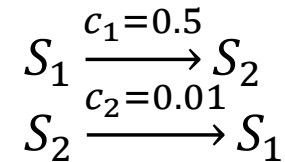


- Gather together in groups
- Specify the state-change vector of each reaction
- Follow the SSA flowchart to:
 - Calculate the propensity function $a_\mu(\mathbf{X})$ of each reaction
 - Calculate the value $a_0(\mathbf{X})$
 - Generate two random uniform numbers (you can [use this website!](#)) and
 - Choose the reaction R_μ that will be applied at each time step and the waiting time τ , that is, the length of the time step before the next reaction can be applied
- Plot the outcome of your simulation (pay attention to times!)

SSA - Python code



- I uploaded in Moodle an implementation of Gillespie's algorithm
 - The code is (almost) general-purpose – you can **implement the support for the missing reactions** as an exercise, just extend the method `_get_h()`
 - The code implements the following example model with a single reversible reaction:



Initial state: $\langle S_1: 100, S_2: 0 \rangle$

- The script also contains some code to **plot the results**
- We will use this implementation as a starting point for the first **laboratory**



First Reaction Method

- The version of SSA we just saw is named "direct method" (DM), because it calculates τ and the index μ directly
- There is another version of the algorithm named First Reaction Method (FRM) which changes the DM as follows:
 - Calculate all propensity functions
 - Calculate a putative time τ_i *for each reaction* R_i according to an exponential distribution with parameter a_i
 - Let μ be the reaction with the smallest putative time τ_i



Why not using the FRM?

- It can be mathematically **proven** that the **FRM** and **DM** are **equivalent** and both produce an **exact realization** of the underlying CME
- However, we have to sample the exponential distribution for all M reactions, which requires to **generate M random numbers**
 - Just two random numbers per step are used for the DM
 - Hi quality pseudo-random numbers are **computationally expensive!**
 - FRM scales poorly with larger models



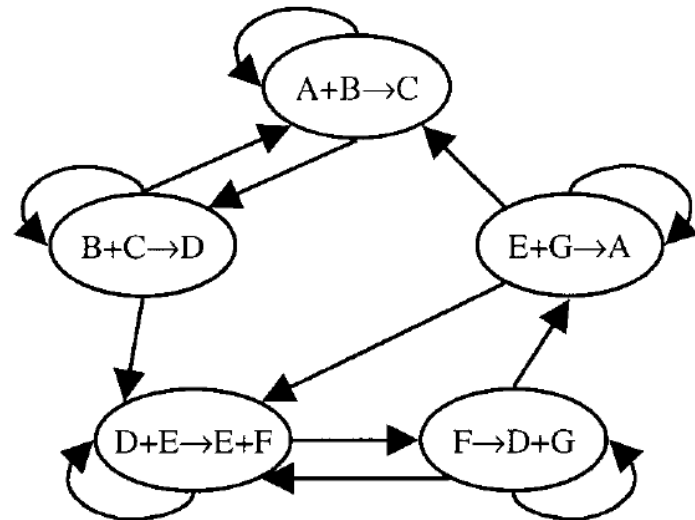
Next Reaction Method

- The **Next Reaction Method** (NRM) was proposed by Gibson and Bruck in 2000 as a "smart" extension of the FRM
 - They noted that, after the firing of a reaction, **only a subset** of the propensities must be updated
 - They proposed to maintain a special data structure (a **dependency graph**) that summarizes **which propensities are affected by the firing of each reaction**
 - They also use another data structure (a priority queue) to **keep the putative times ordered**, and **update only the putative times that were invalidated** by a propensity change
 - [Gibson and Bruck, *J Chem Phys* 2000]
- Let's understand these ideas with a simple example

Dependency graphs



reaction	a_μ	$DependsOn(a_\mu)$	$Affects(\mu)$
$A + B \xrightarrow{k_1} C$	$k_1 \times (\#A) \times (\#B)$	A, B	A, B, C
$B + C \xrightarrow{k_2} D$	$k_2 \times (\#B) \times (\#C)$	B, C	B, C, D
$D + E \xrightarrow{k_3} E + F$	$k_3 \times (\#D) \times (\#E)$	D, E	D, F
$F \xrightarrow{k_4} D + G$	$k_3 \times (\#F)$	F	D, F, G
$E + G \xrightarrow{k_5} A$	$k_5 \times (\#E) \times (\#G)$	E, G	A, E, G



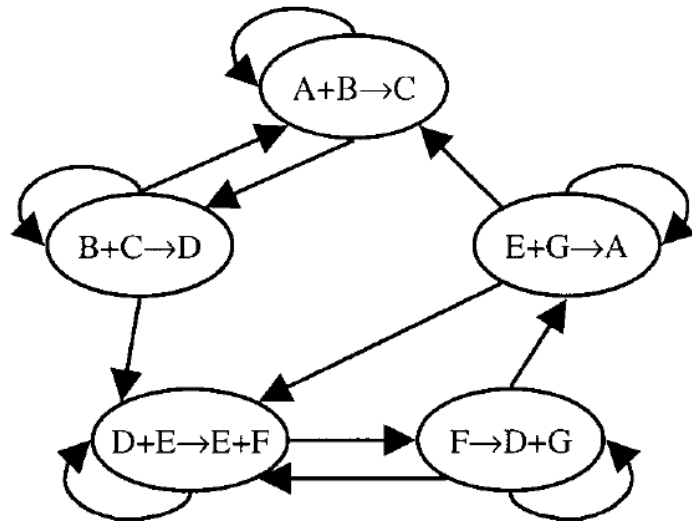
Dependency graphs



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$A + B \xrightarrow{k_1} C$	$k_1 \times (\#A) \times (\#B)$	A, B	A, B, C
$B + C \xrightarrow{k_2} D$	$k_2 \times (\#B) \times (\#C)$	B, C	B, C, D
$D + E \xrightarrow{k_3} E + F$	$k_3 \times (\#D) \times (\#E)$	D, E	D, F
$F \xrightarrow{k_4} D + G$	$k_3 \times (\#F)$	F	D, F, G
$E + G \xrightarrow{k_5} A$	$k_5 \times (\#E) \times (\#G)$	E, G	A, E, G

The first reaction **affects**
all reactions that contain
the **species A, B and C**

Including
itself!

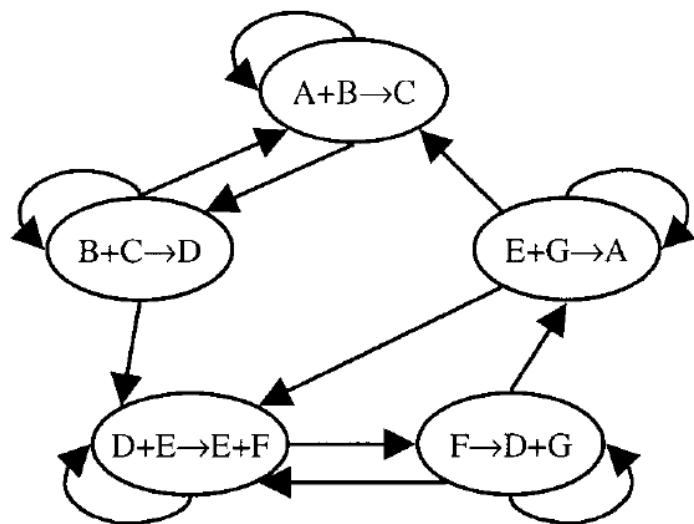


Dependency graphs

reaction	a_μ	$DependsOn(a_\mu)$	$Affects(\mu)$
$A + B \xrightarrow{k_1} C$	$k_1 \times (\#A) \times (\#B)$	A, B	A, B, C
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$D + E \xrightarrow{k_3} E + F$	$k_3 \times (\#D) \times (\#E)$	D, E	D, F
$F \xrightarrow{k_4} D + G$	$k_3 \times (\#F)$	F	D, F, G
$E + G \xrightarrow{k_5} A$	$k_5 \times (\#E) \times (\#G)$	E, G	A, E, G

The first reaction **affects** **all reactions** that contain the **species A, B and C**

Including itself!

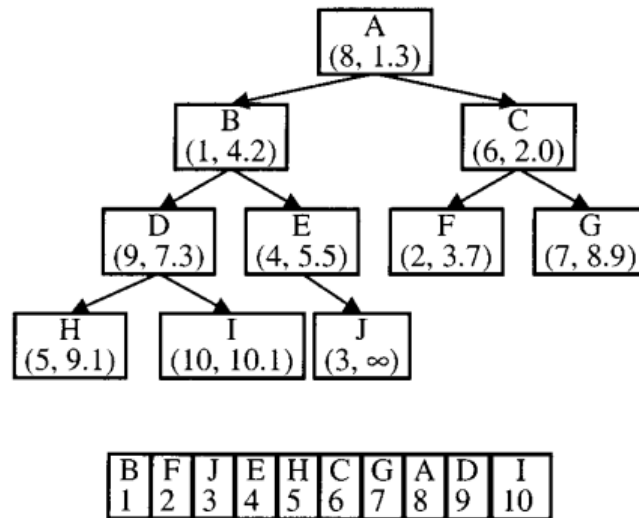


By inspecting the dependency graph, we notice that **each reaction has at most three outgoing arrows** (see, e.g., $B+C \rightarrow D$)

Updating **just 2 or 3 propensities** per step instead of ricalculating ALL propensities provides a nice **speedup**

Priority queues for the putative times

- The smallest putative time is always the root of the tree
- Whenever a **propensity changes**, we **recalculate the waiting time** and only update the **affected part of the tree**



Algorithm 3. *UPDATE*(node n , value new_value)
 Change value of n to new_value
UPDATE_AUX(n)

Algorithm 4. *UPDATE_AUX*(node n)
 If $value(n) < value(parent(n))$
 SWAP n and $parent(n)$
 Update_aux($parent(n)$)
 Else If $value(n) > minimum\ value(children(n))$
 SWAP n and $minimum\ child(n)$
 Update_aux($minimum\ child(n)$)
 Else
 Return



SSA's bottleneck

- Regardless of the SSA version we are using, there are some intrinsic bottlenecks
- Propensity functions are used to evaluate the next reaction R_μ and the time τ before that reaction occurs
 - The smaller the τ , the longer the simulation time
 - The propensity functions are proportional to the product of the number of reactant molecules $\rightarrow$ SSA is slowed down in systems characterized by a high number of molecules

Summary



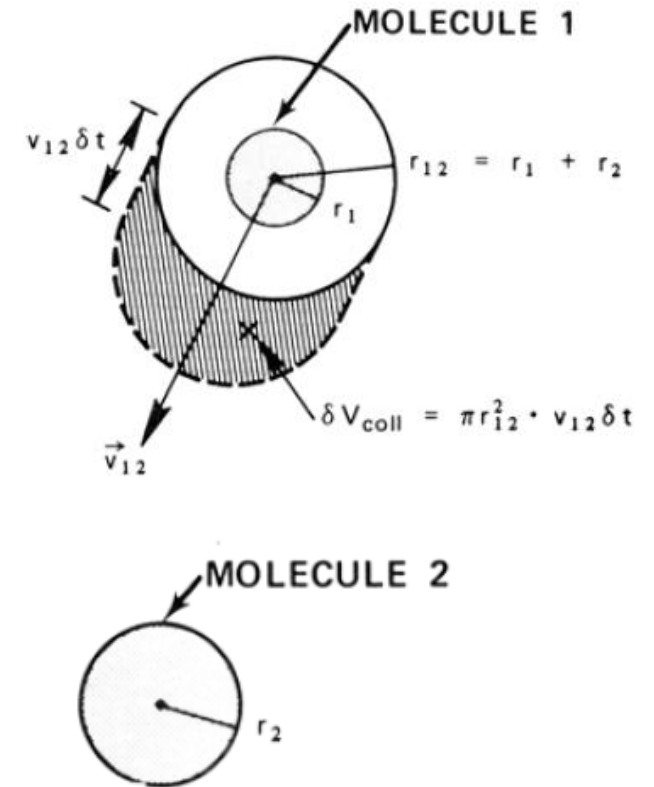
- Complex systems can be investigated by means of modeling and simulation
- We need to figure out the appropriate scale for the model we want to study
- We also need to determine whether deterministic or stochastic approaches should be employed
- When a few molecules are involved, noise plays a relevant role and produces emergent phenomena (e.g., bistability, state switching)
- (Exact) stochastic simulation can be computationally very intensive

The physical basis of stochastic approach

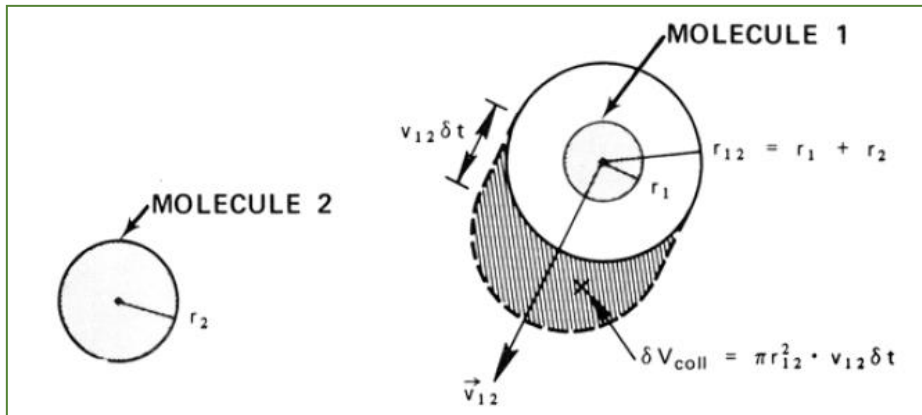


Consider a reaction between one molecule of species S_1 and one molecule of species S_2 , where:

- S_1 and S_2 are hard spheres with radius equal to r_1 and r_2 , respectively
- $r_{12} = r_1 + r_2$ is the center-to-center distance that needs to be reached for a molecule S_1 and a molecule S_2 to collide
- v_{12} is the speed of molecule S_1 relative to molecule S_2
- δV_{coll} represents the “collision volume”: if the center of molecule S_2 happens to lie at time t inside this space, then the two molecules will collide in the next infinitesimal time interval δt
- The “collision volume” is a “cylinder” having a volume equal to $\pi r_{12}^2 v_{12} \delta t$
- If $\delta t \rightarrow 0$ then there is only a negligible probability that a collision between S_1 and S_2 in δt is prevented by an earlier collision of either S_1 or S_2 with another molecule
- Under the assumption of spatial homogeneity, the value $\delta V_{coll}/V$ represents the probability that the center of one molecule of S_2 will lie within δV_{coll}



The physical basis of stochastic approach



For more details, see the papers by D. Gillespie

- Under the assumption of spatial homogeneity, the value $\delta V_{coll}/V$ represents the probability that the center of one molecule of S_2 will lie within δV_{coll}
- Then, the *average probability* of a S_1 - S_2 collision is: $\overline{\delta V_{coll}/V} = V^{-1} \pi r_{12}^2 \overline{v_{12}} \delta t$



The physical basis of stochastic approach

- If, at time t in V , there are X_1 molecules of S_1 and X_2 molecules of S_2 for a total of $X_1 X_2$ distinct combinations, then the *probability* of a S_1 - S_2 collision in the next infinitesimal time interval $(t, t + dt)$ is equal to:

$$\overline{\delta V_{\text{coll}}/V} = V^{-1} \pi r_{12}^2 \overline{v_{12}} \delta t$$

$$X_1 X_2 \overline{V^{-1} \pi r_{12}^2 v_{12}} dt$$

The **red box** corresponds to the «fundamental hypothesis» (average probability), the **green box** corresponds to the stochastic constant, the **blue box** corresponds to the probability of the reaction

$$(8kT/\pi m_{12})^{1/2}$$

Maxwell velocity distribution, where:
 k = Boltzmann's constant
 T = absolute temperature
 $m_{12} = m_1 m_2 / (m_1 + m_2)$ (reduced mass)

The stochastic constant takes into account all the chemical and physical properties of the reaction